

SYNTHESIS PROJECT

An AprilTag-Based Approach to Trajectory Recording and Relocalization for Mobile Robots

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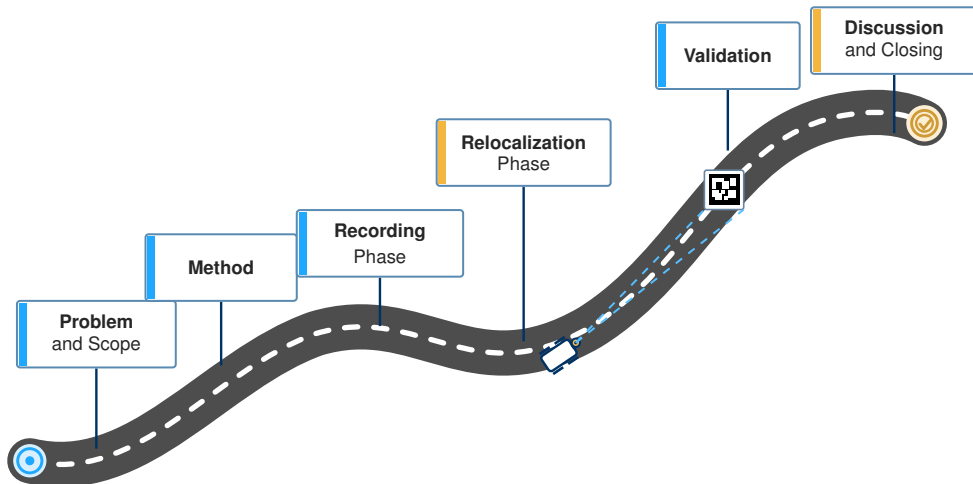
Master's degree in artificial perception and robotics

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Outline



Problem and Scope

For whom

Polytech practical-work students using AgileX LIMO in indoor AprilTag setups.

Why

Need a reproducible workflow to record a path and recover pose relative to it for control-law exercises.

Problem

Recording alone is insufficient because odometry drift breaks trajectory consistency across runs.

Scope

In: offline reconstruction and relocalization outputs (D , ϕ).
Not delivered here: trajectory tracking controller implementation.

Problem and Scope

Scope: control law design is future work

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Method

Recording Phase

Relocalization Phase

Validation

Discussion

Conclusion

Thank you for listening!

Any questions?

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Record, Map, and Relocalize with AprilTags